Learning-Completion-and-Engagement-Analysis-Report

Evidence Based Review of Learner Completion and Engagement

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Analysis tool	Microsoft Power BI

This report consolidates the evidence presented across the three Power BI dashboard pages and translates the findings into business recommendations.

Executive Summary

LearnNaija Academy is reviewing a decline in learner completion and the potential effect on customer retention and corporate renewals. Management reported an overall completion rate of 23 percent, including 19 percent for individual learners and 61 percent for corporate learners. The supplied dataset of 10,000 enrolments presents a different picture: overall completion is approximately 35.70 percent, with almost no difference between corporate and non-corporate learners.

The analysis does not provide strong evidence that device type, corporate status or delayed first login independently explains non-completion. Course category and support-contact patterns show modest differences, but the gaps are not large enough to establish a clear root cause. More importantly, 4,545 enrolments contain at least one identified data inconsistency, limiting confidence in progress and completion measures.

Main recommendation. Do not commit the proposed ₦40 million investment solely on the current evidence. First repair completion and progress measurement, capture the missing behavioural and technical variables, and test interventions through controlled pilots.

Management Context

Management indicator	Reported value	Business implication
Overall course completion	23%	Learner completion is below expectation.
Individual learner completion	19%	The individual segment appears especially weak.
Corporate learner completion	61%	Management believes corporate support improves outcomes.
Early learner drop-off	62%	Many learners are believed to disengage early.
Renewal revenue at risk	₦64 million	Completion problems may affect retention and renewals.
Proposed solution investment	₦40 million	A micro-course and mobile-first response is being considered.

Project Purpose and Scope

The purpose of this capstone was to use stakeholder interviews, customer feedback and the simulator dataset to assess the stated hypotheses about low completion. The analysis covers learner type, first-login delay, device, support contacts, course category, recorded progress and completion. It also reviews whether the dataset is suitable for management decision-making.

Key Questions

- Do corporate learners complete courses at a higher rate than non-corporate learners?
- Does delayed first login reduce completion?
- Do mobile learners complete less often than laptop users?
- Do learners who contact support more frequently complete less often?
- Do completion outcomes differ materially across course categories?
- Is recorded progress reliably associated with completion?
- Can the dataset support a major investment decision?

AS IS Process

The AS IS process describes how the current learning journey and reporting process operate before any improvement is introduced.

Stage	Current activity	Observed concern
1	Learner enrols in a course	Marketing source and motivation are not captured.
2	Learner receives onboarding communication	Onboarding experience and email engagement are absent from the dataset.
3	Learner logs in and watches lessons	Buffering, data cost and lesson duration are not recorded.
4	Learner may contact support	The type, severity and outcome of support are unknown.
5	Platform records progress and completion	Some records conflict with lesson totals and completion status.
6	Management reviews aggregate reports	Management figures do not reconcile with the supplied dataset.

Dashboard 1 Executive Overview

The first dashboard summarizes enrolment volume, completion, progress and the main learner segments. It establishes the baseline used for the hypothesis tests on the second dashboard.

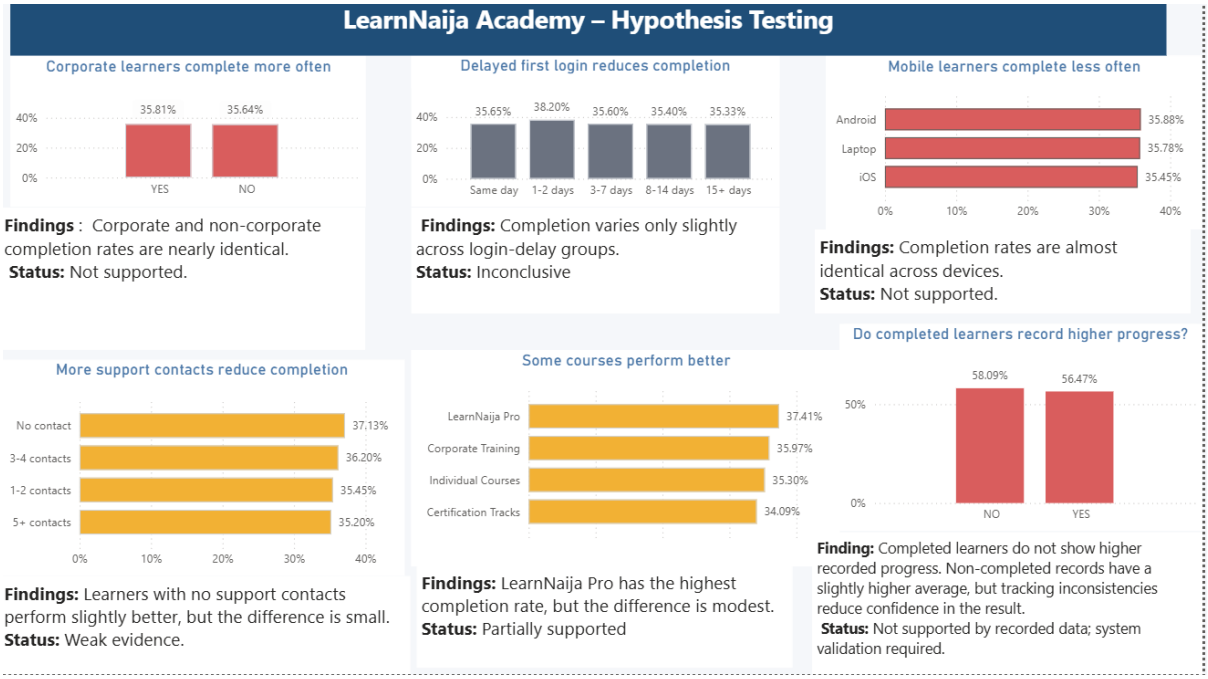


Primary Interpretation

- The dataset contains 10,000 enrolment records.
- Overall completion is approximately 35.70 percent.
- Corporate completion is approximately 35.81 percent, compared with 35.64 percent for non-corporate enrolments.
- The corporate advantage reported by management is not visible in this dataset.
- Average progress should be interpreted carefully because progress and completion fields contain inconsistencies.

Dashboard 2 Hypothesis Testing

The second dashboard tests the main stakeholder hypotheses. Completion rates are calculated as measures and are not displayed as a percentage of the grand total.



Hypothesis Results

Hypothesis	Evidence	Assessment
Corporate learners complete more often	Corporate 35.81%; non-corporate 35.64%.	Not supported
Delayed first login reduces completion	Rates range from about 35.33% to 38.20%, without a consistent decline.	Inconclusive
Mobile learners complete less often	Android 35.88%; laptop 35.78%; iOS 35.45%.	Not supported
More support contacts reduce completion	Differences are small; no-contact learners are slightly higher.	Weak evidence
Some courses perform better	LearnNaija Pro 37.41%; Certification Tracks 34.09%.	Partly supported
Progress is associated with completion	Recorded average progress is slightly lower for completed records and the fields conflict.	Requires validation

Recorded Progress

Completion status	Average recorded progress
Not completed	58.09%
Completed	56.47%

The recorded progress pattern is counterintuitive and conflicts with completion status. This hypothesis cannot be evaluated reliably until the progress and completion rules are validated.

Detailed Findings

Corporate Status

Corporate and non-corporate completion rates differ by only 0.17 percentage points. This does not support the management claim that corporate learners have a substantial completion advantage. The gap between the reported 61 percent corporate rate and the dataset result requires reconciliation.

First Login Delay

First login band	Completion rate
Same day	35.65%
1 to 2 days	38.20%
3 to 7 days	35.60%
8 to 14 days	35.40%
15 days or more	35.33%

Completion is highest in the 1 to 2 day group, but the remaining rates are close. The pattern does not demonstrate that longer login delays cause lower completion.

Primary Device

Device	Completion rate
Android	35.88%
Laptop	35.78%
iOS	35.45%

Completion is almost identical across devices. This does not disprove stakeholder complaints about buffering because the dataset contains no buffering variable.

Detailed Findings Continued

Support Contacts

Support contact band	Completion rate
No contact	37.13%
1 to 2 contacts	35.45%
3 to 4 contacts	36.20%
5 or more contacts	35.20%

Learners with no support contact show slightly higher completion, but the differences are modest. The dataset cannot distinguish a routine question from a serious technical problem, so the result should not be treated as proof that support contact causes non-completion.

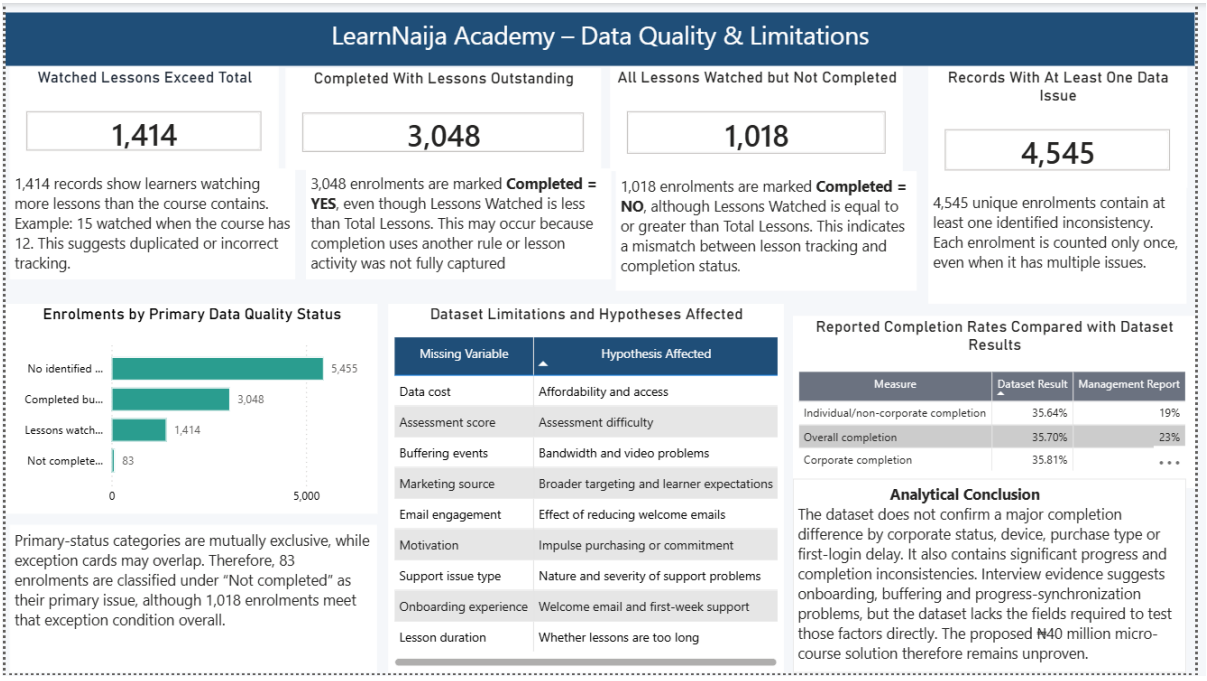
Course Category

Course category	Completion rate
LearnNaija Pro	37.41%
Corporate Training	35.97%
Individual Courses	35.30%
Certification Tracks	34.09%

Course category has a modest relationship with completion. LearnNaija Pro performs best and Certification Tracks performs lowest, a difference of 3.32 percentage points. This supports further course-level investigation, but not an immediate redesign of the entire portfolio.

Dashboard 3 Data Quality and Limitations

The third dashboard assesses whether the data is reliable enough to support the business decision. It compares management figures with dataset results and highlights gaps that prevent several stakeholder statements from being tested.



Data Quality Exceptions

Exception	Records
Lessons watched greater than total lessons	1,414
Marked completed with lessons still outstanding	3,048
Not completed despite all lessons being recorded	1,018
Unique enrolments with at least one identified inconsistency	4,545

The first three exception counts may overlap. The 4,545 figure is the deduplicated number of enrolments with at least one identified inconsistency, representing 45.45 percent of the dataset.

Management Report and Dataset Reconciliation

Measure	Management report	Dataset result	Difference
Overall completion	23.00%	35.70%	+12.70 percentage points
Individual or non-corporate completion	19.00%	35.64%	+16.64 percentage points
Corporate completion	61.00%	35.81%	-25.19 percentage points

The differences are material and could reflect different time periods, populations, definitions, filters or data pipelines. Management should not use the figures interchangeably until the reporting logic and source data are reconciled.

Missing Evidence

Missing variable	Hypothesis or decision affected
Buffering events	Bandwidth and video-performance complaints
Data cost	Affordability and access
Onboarding experience	Welcome email and first-week support
Marketing source	Targeting and learner expectations
Motivation	Impulse purchasing and commitment
Lesson duration	Whether lessons are too long
Support issue type	Nature and severity of learner problems
Assessment score	Assessment difficulty and learning outcomes
Email engagement	Effectiveness of onboarding communication

Root Cause Assessment

The available evidence does not identify one confirmed cause of low completion. Instead, it points to a measurement problem and several plausible operational causes that require additional evidence.

Potential cause	Current evidence	Conclusion
Corporate support model	Corporate and non-corporate rates are almost equal.	Not supported by dataset
Delayed first login	Small and inconsistent differences across login bands.	Inconclusive
Mobile device use	Device rates are nearly identical.	Not supported by dataset
Support burden	No-contact learners perform slightly better, but issue type is missing.	Weak evidence
Course design	A modest spread exists across course categories.	Partly supported
Progress tracking	Progress and completion fields conflict.	Measurement failure
Bandwidth and data cost	Raised by stakeholders but not captured.	Cannot be tested

The strongest immediate root cause is unreliable measurement. Possible learner-experience causes remain hypotheses until the missing variables are captured and tested.

Recommended TO BE Approach

Priority	Recommended action	Expected outcome
1	Define one approved completion rule and reconcile event tracking across devices and reports.	Reliable and consistent KPIs
2	Capture buffering, onboarding engagement, support issue type, lesson duration and assessment outcomes.	Evidence for currently untestable hypotheses
3	Pilot onboarding, short-form content and low-bandwidth delivery with control groups.	Measured causal evidence
4	Scale only interventions that materially improve completion, engagement or retention.	Investment linked to verified value

Implementation Plan

Phase	Timing	Main activities	Success measure
Data repair	Weeks 1 to 4	Agree definitions, investigate exceptions, reconcile management and dataset reports.	One trusted completion KPI
Instrumentation	Weeks 3 to 8	Add missing event and experience fields; establish validation controls.	Required fields captured accurately
Pilot design	Weeks 6 to 10	Select learner groups, define baseline and treatment, agree decision thresholds.	Approved test plan
Controlled pilots	Weeks 10 to 16	Test onboarding, content length and low-bandwidth options.	Measured uplift against control
Investment decision	After pilot review	Compare benefit, cost, retention effect and operational feasibility.	Evidence-based funding decision

Measures for Ongoing Monitoring

- Overall and segment completion rate using the approved definition.
- First-week activation and lesson engagement.
- Buffering frequency and failure rate by device and network condition.
- Support-contact reason, response time and resolution outcome.
- Course-level completion, assessment success and withdrawal reason.
- Renewal, repeat purchase and retention by learner segment.
- Data-quality exception rate and reconciliation status.

Conclusion

The three dashboards provide a consistent message: several stakeholder assumptions are not supported by the current dataset, and the available data cannot reliably explain why learners fail to complete. The most important action is to establish trustworthy measurement before approving a major solution investment. Once the data foundation is repaired, LearnNaija should run controlled pilots and fund only the interventions that demonstrate measurable business and learn